

1. The slow (sometimes called “brute force”) algorithm for finding the shortest common superstring of the strings in set S involves: 1 point
- ☒ Trying all orderings of the strings in S
 - ☐ Finding the longest common substring of the strings in S
 - ☐ Concatenating the strings in S
 - ☐ Iteratively removing strings from S that don't belong in the superstring
2. Which of the following is **not** a true statement about the slow (brute force) shortest common superstring algorithm. 1 point
- ☐ The amount of time it takes grows with the factorial of the number of input strings
 - ☐ It might collapse repetitive portions of the genome
 - ☒ The superstring returned might be longer than the shortest possible one
3. Which of the following is **not** a true statement about the greedy shortest common superstring formulation of the assembly problem? 1 point
- ☒ The amount of time it takes grows with the factorial of the number of input strings
 - ☐ The superstring returned might be longer than the shortest possible one
 - ☐ It might collapse repetitive portions of the genome
4. True or false: an Eulerian walk is a way of moving through a graph such that each node is visited exactly once 1 point
- ☐ True
 - ☒ False
5. If the genome is repetitive and we try to use the De Bruijn Graph/Eulerian Path method for assembling it, we might find that: 1 point
- ☒ There is more than one Eulerian path
 - ☐ The genome “spelled out” along the Eulerian path is not a superstring of the reads
 - ☐ The De Bruijn graph breaks into pieces
6. In a De Bruijn assembly graph for given k, there is one edge per 1 point
- ☐ read
 - ☐ k-1-mer
 - ☒ k-mer
 - ☐ genome

7. Which of the following does not help with the problem of assembling repetitive genomes:

1 point

- ☐ Paired-end reads
- ☐ Longer reads
- ☒ Increasing minimum required overlap length for the overlap graph